Active InferAnt Stream 010.1 ~ plain_text for active_inference: Modeling, L

ActiveInferAntStream | plain_text for active_inference: Modeling, Learning, and Exploration | 1:57:53

All right, we're live.

Welcome everyone.

It's February 7th, 2025.

And we are in ActiveInfra Ant Stream 11.

Looking forward to a lot of feedback, interactivity, and plain text fun and learning.

All the pushes during the stream are going to be going to the repo doxology slash cognitive.

And let's get right into it.

All right, so previously on ActiveInfra Ant Streams, we've heavily featured the cursor development environment tool.

And that's going to continue to be used as well as seen a little bit with CodeViz.

And that's going to come into play again.

And in this stream, for plain text, surprise, education, and delight, we're going to be working a lot with Obsidian.

So Obsidian at the repo or website obsidian.md is a plain text note linking software, open source.

And it is going to facilitate us doing some truly awesome things with plain text repos.

And playing back and forth between Cursor and Obsidian is going to go really far.

So here in the Obsidian window, on the left side, we're looking at the same file that on GitHub can be found here.

And in Cursor can be found here.

So we have two local versions of Infra Ant Stream 10.md.

So if we delete something, like starting with a GitHub push, save it in Cursor, that's going to be gone in Obsidian.

Obsidian uses a linking style where two brackets open up the ability to make a linker.

So that is going to be reflected visually in this graph version.

So before we go too deep in, this is some incipient, exciting, relevant development work.

It's semi-structured, so don't let the graph and the weave make too much light of the fact that if we zoom out,

of course there's more to paint into than not.

And looking forward to people's collaboration and contributions.

You can head over to the repo while the stream's happening or afterwards.

Or if you're watching live, definitely put some funny questions and ideas into the live chat.

And we'll bring them into the repo and that will weave that way.

Big picture, we're going to be in some ways that were not really even possible a few years ago,

integrating theoretical foundations, writing them on the fly, and connecting with practical implementations.

So we can delete that and then we can see, okay, now we just updated a plain text file.

So let's just make that push just for fun.

Okay, where do you even begin?

So welcome an overview of Obsidian usage.

So this could have been typed in manually like this.

Two brackets, Obsidian usage.

When we click on that link, it's like Wikipedia.

We head over to this article.

And there's going to be links in the article.

The ones that are brightened are the ones where there is a page.

And the ones that are dim, there's no page yet.

And so if we click on that, it makes that page at the top level of the repo.

So we don't need that at this point.

But let's look a little bit at Obsidian usage.

And this is a markdown format.

Everything is these plain text markdown files.

Okay, so file names link directly.

File name, vertical line, alias uses custom text.

If you want it to say something different than the file name.

And here we have some machine readable information.

And then over on the graph side, we can filter down and try to see what's in the neighborhood of Obsidian usage.

There's going to be a lot of active inference generative models in this stream as well.

But first, I just wanted to start this way.

So here we are.

And when we have our cursor there, we get this colorful highlight of Obsidian usage.

And everything that it's linked to in the plain text gets rendered as an edge here.

So we can drag around that, play with that.

Okay, Obsidian usage, plain text benefits.

Let's check out plain text benefits.

Wow.

Version control, knowledge management, machine readability, research benefits.

All these different benefits.

Which cursor wrote entirely for me?

And we'll get to how and where that happened.

Okay.

Back to the top level and let's carry on with the stream.

So, preview streams, we've been using different synthetic intelligence and live chat methodologies for having a great time learning about what other than active inference and the free energy principle.

So, let's check out those articles.

Again, these were written entirely using cursor.

So, what that looked like was, let's head over to knowledge base, mathematics.

We can go over to some active inference article.

So, here's the plain text article.

This is it in Obsidian.

So, here we have everything from software examples to it could be anything.

If we want to use AI to update this active inference article, bring it in with control I to the composer mode.

And then we can say add fantastic, comprehensive, relevant length, and structured typo new sections to the active inference article.

So, now if we head over to active inference.

We see it's connected to a whole variety of different terms and topics and ideas.

So, very cool how also by embedding a docs section, there's a lot of Obsidian style.

Also, just picking up from context.

This is, so here is Claude 3.5 Sonnet 2024-1022.

I've added four major new sections to the active inference article.

So, here in the green are these new sections that just got added.

Again, you'll remember from earlier inference streams where anything above like 200 lines of code was getting cut off.

Now we're into the 1400 lines of code.

So, that was a pretty simple edit.

We'll accept it.

But it changes the graph and you can watch cursor modifying files.

And then it's popping up and the graph topology is evolving.

So, here's free energy principle.

And these gray nodes are all hypothetical links.

So, they're linked kind of in a preemptive way.

We can turn on to just existing files if we want to look at the more dense weaving of,

okay, well, how about variational calculus?

This comes up like all the time.

All right.

Here we have different math, different examples, different applications of variational inference.

And these on a line-by-line basis can be vetted in different ways and researched by different individuals.

So, if it were like tags, I'll add a tag, whatever it is.

And then that can be versioned.

And if people disagree, they can use the other version of the plain text file.

File.

Hey, Andrew.

Yeah, there's going to be some really fun applications here.

Okay.

So, let's now go to the agenda.

So, free energy principle article, just briefly.

Also, super long article with all of this machine-readable information.

We're going to explore a bunch of fun plain text and that change.

So, here's where we are.

Here's stream 10 connected to act-dimpf and FEP.

And then, when we delete this line, it disconnects.

Briefly, going to go through the doc part of the repo to show a little bit of the metastructure on the cognitive repo.

And then, we're going to get into POMDPs, partially observable, Markov decision processes, our favorite, kind of Drosophila, model creature for active inference, and ANTs and bioferms and more.

Oh, brother.

So, here we go.

First, into the repo.

This config folder looks empty.

Could have been...

It's a simulation config.

We'll leave it there for now.

Maybe it's not even important.

Okay.

Here's a documentation roadmap.

Here it is in cursor.

Here's documentation in Obsidian.

And a readme for the documentation.

Some of this might seem like overkill.

However, as long as it's accurate and gives linking valid structure, this is incredibly effective with cursor and Composer.

So, first, API.

That's not really being used at this point, but there's a lot of discussion around agent APIs.

And so, having these comprehensively linked versions of files in plain text, super powerful, but not really going to come into play here.

Concepts.

This is just documentation for concepts.

So, that are relevant for the package, but not scientific concepts per se.

More like concepts that are relevant for the documentation at a meta level.

So, that when we say, well, we're really interested in machine readability.

That kind of semantics is going to get across just fine.

But if we provide a ton of linking so that machine readability is a full knowledge node, then it can be even more effective.

And it can have these mermaid diagrams.

So, then, when we're giving a talk and we're going, well, free energy principle, we want to have reproducibility and research integrity.

And that's why we're studying this cognitive phenomena with machine readability.

Those are just paths just around the horn.

6, 4, 3, double play.

Examples.

These are some examples of documentation.

Again, it's really not the most exciting afternoon reading per se, but it gets really far with helping cursor guides.

These are guides for different micro standard operating protocols for cursor and agents.

And these can be taken out to extreme levels of intensity.

So, again, it could be followed by a person or it can just be used with these absurd levels of hierarchical dispatch.

Research.

And it can be used with this augmented niche.

And it can be used also to list other autonomous AI tooling.

So, it could be like search literature database or drive numerical belief from natural language.

Any of these things that people have been exploring for years and also using Obsidian.

So, let's just make that file longer.

Obsidian usage.

Make this Obsidian use file much comprehensively longer and with some relevant linked files, comma, make them fully.

So, this is using the agent mode, which right now only exists for the Claude family, I believe.

However, we may expect and prefer not so long when many of these models from the OpenAI family, from the Google, from various other labs are useful.

And integration with local language models, all these kinds of things are more and more useful and possible every day.

So, I'm in the slow usage tier.

I'm in the slow lane right now because of what has been happening over the last several days.

We'll just let that cook.

Maybe it'll make some changes while we're looking at it, but let's carry on for now.

Okay.

That was just the documentation folder.

Structured approaches for building documentation and advancing documentation from any current state and forensically reconstructing documentation.

These are some truly incredible opportunities.

Okay.

Now we get into the knowledge base itself.

Things is where we're going to see the active inference models, but I want to walk through these top sections because they're really fun.

Okay.

Okay.

Let's go over to systems theory.

So, here's systems theory.

We can see already that it's going to link to cognitive phenomena.

It's going to link to swarm intelligence.

It's going to link to an index file.

So, here's a short linker article on systems theory.

And we can follow some of these links.

Like, how about information theory?

And we can follow this.

All written iteratively through cursor.

And every single line could be checked.

There could be a person.

There could be any number of other approaches.

Here's what the Obsidian update is doing.

So, it's like, okay.

And it gets it that we're within an Obsidian repo.

So, here now it's updating the Obsidian documentation for itself.

There's probably some points of diminishing return, but it just wrote several hundred lines.

And then now it's going, okay.

Well, let's make a new linked file for network analysis.

So, here it created this whole new file for network analysis.

Very, of course, meta, since this is what's happening over here.

And that can be done in multiple different maneuvers by one press here.

So, these are new articles that it's writing.

Okay.

Okay.

Back to the knowledge base.

Here are the different areas.

Citations, I've left empty.

Citation format and bringing in full text.

That's a whole other topic.

Not for this repo this time.

There's agents, which is documentation on specific agents.

So, here it's a full article.

O and S space.

O I added just manually as an example of one that we can build right now.

S space.

This is a whole article with linkages on S.

Latent state or hidden state.

So, we can have mega connections with the active inference ontology.

And this helps entrench different understandings.

So, that one is like A matrix.

Here's exactly what the structure is.

This is repeat giving multiple explanations.

So, it is wild times.

For years, it's like, well, what really is the A matrix?

Or what is the A matrix?

Or how are different ways to write it?

Or what letter should we use?

Or what's the most interesting?

It's like, we're kind of in a position now where it's more about,

the underlying entrenchment of understanding,

plus perhaps we could say the crystallized or classically modified understanding of each of these variables.

And then, the more fluid situational and contextual deployment of these structures.

So, let's have the O space article get written.

So, in agents, generic POMDP, O space.

Here, write the O space article fully about the observation space.

Okay.

Okay.

So, now we're looking at this graphical relational matrix describing the knowledge network describing a POMDP structure.

So, here in the POMDP structure, it's like, hey, when I say POMDP, these are the components that I mean.

So, O space was empty.

It's still getting written.

But we're going to see the graph structure change.

Let's see if we can catch it live changing.

Okay.

Now, O space has been written.

234 lines.

So, it relates to all these other topics.

So, there's like a two-way mapping.

Okay.

I guess observation, it does relate to this concept of observation structure.

Should we have an extensive human-to-human experience debating whether or not observation structure should be its own page or this page or that page or the first slide in your presentation or this or that?

It's like, maybe not.

But having it as a hypothetical node that we can continue to unpack iteratively and crawl through is another level and type of post-Wikipedia knowledge engineering.

But maybe bringing back the relevance for Wiki and plain-text collaborative design tools as well.

So, here we are in O.

Now, we can see the structure of the POMDP fleshing out.

So, we were looking through the generic POMDP documentations.

Like, levels of documentation that I don't even know have been seen for some of these topics.

Because we can always just go into a given matrix.

Vastly expand the comprehensive article on B matrix.

These are the kinds of thrilling articles that I'm sure many of the listeners and viewers and live chatters love to write in their spare time.

But now we are getting assistance with writing these fun articles.

Let's look at the B matrix.

And it's going to be evolving its topology.

So, that's the POMDP.

We also have continuous time agent that we'll get to.

Citations.

Again, there's none at this point.

Cognitive.

Cognitive.

There's a variety of cognitive mechanisms or cognitive phenomena.

This is very similar to the work we did in the Atlas project at CogSec related to pattern languages for cognitive phenomena.

So, here in the cognitive phenomena, here's the improved extended article on the B matrix.

So, there's more connections to computational topics.

More connections to how we like to see it implemented.

More math foundations.

And then someone can spin it however they want.

Like, you know, the classic.

Write the who's on first to understand the B matrix from a perspective where I don't know much math past middle school.

So, custom generative multi-language, multi-modal educational materials is in the game.

So, updates.

Push it to GitHub.

So, it interpreted that perhaps because of the previous context with writing a new knowledge format.

So, it's going to write a new document called B matrix basics.

So, we'll see what it comes up with there.

Cognitive phenomena.

Let's see it in cursor.

Cognitive phenomena.

Here, we'll use control K.

So, we can do inline coding.

We'll say vastly add phenomena to this list.

Here's B matrix basics.

Here's B matrix basics.

Here's B matrix basics.

Here's B matrix basics.

Here's B matrix basics.

Here's B matrix basics.

Here's B matrix economics is bound to make known to be known to

pages for, others we don't. But you could imagine iteratively applying prompts.

Where there is a page with no information, please write that

comprehensively, vastly, excellently, professionally, etc. So that'll update

cognitive phenomena. But back to the cognitive phenomena folder, we have all

these kind of, here's an empty one, but all of these on-demand articles. So almost surely trained on a variety of articles about ants and collective behavior, for example. But at the same time, this is coming up with machine-readable schema and ontologies.

That you could have all the myrmecologists in the world in a room, and what they know of might come up with a very high correlation of these terms.

And then it's always possible to just add more terms. If someone says, well, it's like, well, they really came up with a new term that should go here.

And then that's where you can just add the new term directly, and then build on the page. And then you have the page template, and then it links to the documentation, and so on.

So cognitive phenomena, just a bunch of times I iteratively applied a few prompts, just to get it to make these kind of wiki style articles on all these different topics, whether more as a linker, which is good for making like a beachhead into a new epistemic frontier, or a more advanced article that has visualizations.

I think for the cognitive phenomena one, I was just interested to get it to link as much as possible, which is why it has this list format.

But then you say, for every single bullet point, expand that section so that it has three sentences.

And then next prompt, every sentence, split it to reify the main point with an example.

That's the cognitive section.

Okay, mathematics. So here, it's pretty fun because the LaTeX rendering is so nice.

Nice.

We can look at belief updating, math, code, connections.

How about compute expected free energy?

Well, here's the definition of expected free energy in terms of epistemic value and pragmatic value.

So we have this more wiki documentation page.

But then when in the active inference page, it says it's going to use expected free energy.

And then that is linked strongly to the expected free energy page.

And here's what we expect from expected free energy.

So then when I prompt somewhere else, something like make an active inference model or make an active inference POMDP model of this or that situation.

These highways and byways are right there to trace it back to where we have a human interpretable versionable document.

So again, the era of like, what is the best way to convey expected free energy?

It's like, after what's happened in the last 48 hours, in my experience, I would say in plain text, accurately, coherently, professionally, etc.

And then with this generative layer around it, that is able to recognize the query and then connect it to what's relevant to generate.

But if we just dip our ladle into a given LLM and say what is expected free energy, it might give us a totally valid response.

It might give us a semi-memorized response from a paper.

It might come to a emergent synthetic definition from several papers.

Finding a semantic interpolation might be a better version.

So it's not that anything else that's not what I'm showing here is ineffective.

It's that if someone's like, I like this notation system or this should be included, we can just very modularly put that right into the plain text.

So then if we want it explained and say, explain this for me as a enthusiast of topic X, maybe it'll make a new file.

File it, just see if it gives a funny shot. Just, you know, what does it even think these symbols mean?

I don't know. All of us are learning.

But again, these are linked topics.

And so I was just studying and learning going, okay, well, okay, so free energy.

Okay.

We can ask, it could be in here.

Add a section conveying at a big technical picture level, what makes something a free energy?

Who hasn't had that question?

So we just keep adding information.

So a B matrix enthusiast.

Whoa.

Okay.

Let's look at the B matrix enthusiast briefly.

That is in the generic POMDP guides as it basically should be.

Here's the basics.

Here's the who's on first style blog, not dialogue.

And then here's the B matrix enthusiast.

If you've ever been fascinated by how things change over time, whether it's weather patterns, stock markets, or even your favorite video game characters movement, you're already thinking about state transitions.

The B matrix is a powerful tool that helps us understand and predict these changes.

Why it's cool.

So we could make this is kind of real ail

files comprehensively, fully, and lightheartedly.

It probably doesn't even need the accurate information to be also in the Obsidian repo,

but it's like, if this is the stuff that we've already looked at, then it definitely doesn't hurt.

And then it's on the cognitive repo.

Anyone can just download this repo, as well as make their own, all of this, open source, great times,

and then supercharge their own LLM use, so that if we say, make a who's on first dialogue about an elephant

that is using a B matrix to build a bridge, but if we already have elephant, B matrix, build a bridge,

who's on first as their own nodes in a graph, then it's just going to be so much better and so much more reproducible.

Observation space.

It's like, how cool and fun.

Okay, back to the repo, then we'll get through these guides and into the simulation section.

So again, these are like on-demand learning articles

that can't be necessarily taken at 100% accuracy.

However, the same could be said for any document.

And these have the ability to just interactively make diagrams, links.

So for example, the path integral work.

This is some recent work in active inference and FEP that no one is born doing.

So how can we learn this?

Well, there's going into a search engine and trying to find documents based upon backlinking or semantic similarity,

and then reading the primary words that were on that page.

There's trying to find a textbook.

Maybe you have physical access.

Maybe you have digital access.

It's like, or we can kind of spin out a neighborhood like I did several times next to the path integral.

So connect path integral to the FEP.

And then, okay, well, it's in plain text here, but I think if it were

clearly linked there.

Just little nano edit.

I didn't even have to know about the path integral.

Didn't have to write the paper, but just that's a little modular edit

that changes the topology of the network

that helps everyone learn and apply all these topics.

So it's like, or what is an action principle?

These are just things that can be asked in.

So here's the O space enthusiast.

Okay, it wrote A, S, and O.

Here's what makes something a free energy.

So interestingly, this is a common error.

It also deletes a bunch.

Rest of content remains unchanged.

So we'll reject that edit.

But some of these things are just little, little road bumps on the speed run.

Here's the cognitive phenomena.

We said add phenomena to the list.

Okay.

Again, here's a long edit that it, the control K mode does that sometimes.

So look for that.

But these ones, we just added dozens of articles.

And then we can have an iterative prompt or an iterative script that says,

go to cognitive phenomena.

Check to see if the linked file is comprehensively written.

Make it better.

Use an internet search or use your local information thoughtfully,

whatever it happens to be.
And you're just expanding your knowledge base,
which you can start from scratch,
or you could pull what is in this repo already.
Okay.
Okay.
So that's the math learning component.
How is, how is the Taylor series related to the continuous time agents operation in generalized coordinates?
Also something that many have wondered.
These are specific questions that it's like, okay, that was a question about Taylor series expansion, generalized coordinates, continuous time active inference agents.
Those are on one hand, big lifts initially, but on the other hand, they fit so nicely into our working memory and certainly into the context window of these synthetic intelligence artifacts.
So we can ask those questions either directly.
Let's ask that exact question in a new composer window.
See what happens.
How does a.
How does.
How does.
How does it continuous time agents.
In.
Active inference.
Use.
Taylor series.
Expansion.

For.
Navigation and intelligence.
In.
Generalized coordinates.
That question could also just be typed into or spoken to any other LLM interface.
But here, first off, we could be working totally offline and locally.
With a local LLM and these plain text open source files.
Or we can be using.
What.
Whatever it is that's happening here.
So.

Again, just like.

What are we uncertain about?

What are we curious about?

What would be learning and on the edge for us?

And then.

Metacognition.

We.

State that.

That's an action coming from us.

And then.

Write it.

Don't just let that uncertainty dissipate.

And then.

Like.

Okay.

I have this.

Reflex.

Like.

It looks hard.

This looks like it's going to be.

Hard.

To read and learn.

So.

At that point.

It's like.

Is this a lift.

That I'm ready to make.

At this nano moment.

Am I ready to.

Read about this.

Or is it like.

Good enough to know that.

There is an answer.

And maybe dump it into a text file.

For later.

Or maybe.

I'm still struggling with this.

Can it be made into.

A who's on first style.

Dialogue.

Etc.

So.

That's the math.

And that's the knowledge base part.

So this could.

Contain.

Summaries.

Or.

Semantic embeddings.

Or full texts.

For different citations.

And different resources.

Here.

I just did totally.

Ab initio.

From nothing.

Of all of these.

Articles.

Related to.

Fun.

Topics.

Okay.

Moving on past the knowledge base.

Here we get into the SRC.

Or source.

File.

Here.

Is where we can start to explore.

Reliable.

Dis.

Hyphen.

Intermediated.

Calling.

Of abstractions.

For operational approaches.

To active inference systems engineering.

So this is a dispatch.

It's kind of like.

Calling operator.

Let's see it in cursor.

Dispatcher.

Okay.

Here.

We're looking at the active inference dispatcher.

First.

Let's just look over in the graphical view.

What's the neighborhood of the active inference dispatcher?

So these are just some related.

Topics.

Okay.

What the active inference dispatcher does.

Is it picks up.

At the.

Linguistic semantics.

Of the active inference ontology.

And then.

Creates runners.

Through dispatch.

That send off.

To implementational methods.

So if we say.

I want to have an agent.

That uses variational free energy.

On perception.

For updated sense making.

And then.

You can imagine.

Whether we ask.

Or state.

Or it asks us.

It's like.

Okay.

Let's do.

A binary state space.

So it could be zeros.

Or ones.

Coming in for observations.

And then.

Later.

We want to add.

Or change that.

To be in a continuous state space.

Like between zero.

And one.

But also.

For those continuous values.

We would still want.

To be able to use.

The semantic dispatch.

Of variational free energy.

But under the hood.

It's doing a different kind.

Of implementation.

Because of it.

For example.

Being continuous versus discrete.

So that's what.

This dispatch.

Intermediation.

Is doing.

And similarly.

Like for the bio firm.

This is a canvas.

This is a nice feature.

Of obsidian.

Take whole articles.

And lay them out.

Like this.

As well as link them.

To non-articles.

Similarly.

For the bio firm.

In the dispatch.

Here.

Is kind of like.

An interface.

Between.

The high level.

Semantics.

Qualitative.

Linguistic.

And.

Specific.

Opinionations.

And routes.

For lower level.

Implementations.

Someone.

Someone says.

I want a.

Red.

Adaptive.

Funny.

Sedan.

Someone doesn't need to say.

What is the chemistry.

Of the red paint.

Such that it.

Has this.

Wavelength of light.

Under.

It's just like.

Well.

We could list out.

Multiple options.

Or one option.

Or.

You could just.

Hard code and say.

Oh.

When people say red.

They mean.

This.

Exact.

Kind of.

Iron oxide.

You could.

You might not be wrong.

In a given situation.

You might not be wrong.

In a whole variety of situations.

But.

It's not.

The most general.

Systems design.

Approach.

So.

These.

SRC.

Files.

Are like.

A repo.

Of.

Tools.

That.

We want to use.

For different.

Kinds of.

Methods.

That come into play.

Like.

Matrix operations.

So then.

There can be.

A matrix operation.

Like.

Transpose matrix.

And let's just say.

That there's two methods.

At our disposal.

And one of them.

Works well.

For small matrices.

And one.

If it's a big.

One.

So then.

In the.

Outside of the SRC folder.

At the semantic level.

Someone can.

Say.

Transpose this matrix.

At this point.

In the program.

The.

Program's logic.

And then.

The dispatcher.

Sends that.

To whichever.

Implementation.

Is most relevant.

Given some other.

Criteria.

So.

That's a little bit.

Of a stub.

Folder.

It's not a ton.

In there.

But the.

The dispatch.

Concept.

Is going to be.

Critical.

And it's.

An example.

Of something.

That's enabled.

By the active.

Inference.

Ontology.

Okay.

Templates.

Agent template.

Belief templates.

These.

Allow a variety.

Of things.

Again.

They're just.

Examples of templates.

But.

Since the nodes.

In obsidian.

Being.

Plain text files.

Can also be.

Anything.

Templates.

Can be.

Wildly useful.

Because then.

They're used.

For so many things.

Okay.

Now let's get into.

Some of the code sections.

For tests.

All right.

So.

Obsidian.

Maybe there's some way.

To configure it.

Differently.

I just don't have.

Set up right now.

But.

It doesn't display.

Python files.

So here.

We can see that.

In the.

Tests folder.

And in the.

The github repo.

The.

All the files are there.

But obsidian focuses on the markdown.

So pros and cons to that.

But.

It's all good.

Um.

Let's.

Uh.

Open it.

And see what this does.

Okay.

Okay.

This is a test suite.

Okay.

Made some errors.

Start a new.

Compose window.

And while it's fixing that.

Let's look at the outputs.

So these are some.

Package level.

Tests.

Just.

Doing some.

Matrix visualizations.
Different kinds.
Um.
Of.
Simple visualizations.
Analyses.
So.
Not.
Not.
Super complex.
But.
The.
Testing framework.
Is actually slightly more complex than that.
So it can be used in a variety of ways.
All right.
Now we get into the things.
This is.
Exciting because.
It.
It.
It.
Also.
Is pleasantly surprising.
Truly how rapidly.
And absurdly.
Some of these programming capacities.
Are advancing.
So I'm going to start.
With possibly the most useful.
Of the things here.
Which is the generic.
POMDP.
While it's fixing the.
Tests over there.
Also.
In the.
Settings.

There's.

YOLO.

YOLO.

Mode.

Where you can.

Allow.

The commands.

To be run.

So.

Here.

It's both.

Able to propose.

And then.

Run.

And then.

Get.

The terminal.

Feedback.

And then.

Continue to iterate.

In this closed loop.

So.

The agent mode.

Is like.

An order of magnitude.

Or more.

Faster.

And now.

It finished.

All the tests.

Great.

Here's what we did.

But.

That might have taken me.

Several.

Key presses.

I know.

I know.

Several key presses.

To get.

To even that.

And maybe.

I would have said.

Something in between.

That would have stymied it.

You know.

Or maybe.

There was a place.

Where it gets itself.

Into a cold exact.

And I could have come into play.

But.

Suffice to say.

In this situation.

For.

Doing.

A.

Hierarchical debug.

On the testing.

Infrastructure.

For a whole package.

I would not.

Have been able to do that.

With just one click.

Even in cursor.

A couple of weeks ago.

Probably.

Okay.

Let's go to.

The generic POMDP.

So this is an example.

Of using generalized notation notation.

GNN.

For specification.

Rendering.

Implementation.

Execution.

Analysis.

Visualization.

Of active inference.

Generative models.

This is.

Building.

In a very fun way.

On the work.

With Jakob.

From several years ago.

And a lot of contributions.

That.

In questions.

People have had.

Since.

For making these.

Obvious.

Either manually crafted.

Or LLM.

Based transformations.

Of specifications.

Into.

Plain text formats.

That we can then.

Render.

Different kinds of models from.

So.

Here.

Up top.

The lower parts.

Are.

Some of them.

Are.

Are.

Are.

Are.

At this.

Stage.

In this.

Version.

That I'm.

Putting out.

Others.

Are.

Used.

To specify.

Visualization.

So.

This is.

Kind of.

Like.

A whole.

Generative.

Model.

Level.

Config.

Like.

Learning.

Parameters.

Inference.

Parameters.

But.

The minimal.

Config.

Stuff.

Is.

Up.

At the top.

Which.

Is.

The state.

Spaces.

So.

Here.

We have.

And this.

This gets back to our.

O.

O.

Space.

Enjoyer.

B.

Matrix.

Enthusiast.

And so on.

We.

Are.

Stating.

The.

Number.

Of.

Discrete.

Options.

For.

Observations.

O.

Space.

The.

Number.

Of.

Discrete.

Latent.

Spaces.

How.

Many.

Actions.

At each.

Time.

Step.

The agent.

Has.

How.

Many.

Total.

Time.

Steps.

The simulation.

Is.

Going.

To run.

For.

The.

Planning.

Horizon.

For.

The simulation.

Which.

Is.

How.

Many.

Timescales.

Is.

The agent.

Going.

To.

Combinatorically.

Do.

Policy.

Proposals.

For.

And.

Some.

Other.

Features.

So.

Where.

This.

Is.

Going.

To go.

And.

Then.

We'll.

Run.

It.

Is.

Here's.

The.

The.

Plots.

So.

Here's.

An.

Overview.

Of.

Our.

Agents.

Here's.

A.

Five.

By.

Four.

A.

Matrix.

Up.

On.

The.

Top.

Left.

Here's.

Just.

One.

Action.

From.

The.

Matrix.

But.

We.

Can.

Look.

At.

The.

Whole.

B.

Matrix.

Three.

Actions.

These.

Are.

The.

The.

B.

Matrixes.

For.

Each.

Of.

Those.

Three.

Actions.

Pick.

Like.

A.

Slice.

See.

Just.

In.

This.

Visualization.

It's.

Through.

Time.

Like.

Time.

Step.

Zero.

One.

Two.

Three.

Four.

Because.

I.

Did.

Have.

Preference.

Learning.

Fully.

Operative.

And.

Wanted.

To.

Prune.

It.

Back.

To.

Simplify.

It.

Just.

A.

Little.

Bit.

So.

Now.

It.

Can.

Graph.

See.

Preferences.

Through.

Time.

Constraints.

Costs.

But.

It's.

Fixed.

Through.

Time.

D.

Is.

Prior.

On.

Those.

Four.

Hidden.

States.

So.

That's.

Flat.

And.

Different.

Outputs.

From.

Different.

Belief.

Updating.

Through.

The.

Ten.

The.

Action.

Distributions.

But.

The.

Visualization.

That.

Is.

New.

Even.

To.

How.

I've.

Seen.

What.

It.

Could.

Output.

In.

The.

Last.

Several.

Weeks.

Is.

Here.

We're.

We're.

Looking.

At.

Three.

Affordances.

Per.

Time.

Step.

And.

Then.

We.

Have.

The.

Pragmatic.

And.

Epistemic.

Components.

Which.

Are.

The.

The.

The.

Expected.

Free.

Energy.

Hashtag.

Look.

At.

That.

Article.

Playing.

Out.

Through.

Each.

Of.

The.

Time.

Steps.

So.

Here's.

A.

Four.

Planning.

Horizon.

And.

This.

Is.

Like.

Do you.

Go.

Low.

Medium.

Or.

High.

And.

Then.

Here's.

Low.

Low.

Low.

Low.

Here's.

Low.

Medium.

High.

Low.

So.

Let's.

Run.

So.

In other words.

This.

Is.

The.

Whole.

Policy.

Tree.

Rollout.

For.

A.

Generic.

POMDP.

Planner.

Just.

From.

Using.

Generalized.

Notation.

Notation.

In.

A.

Well.

Structured.

Niche.

With.

Some.

Of.

These.

Open.

Source.

Components.

Not.

Even.

Using.

PyMDP.

Or.

Any.

Other.

Kind.

Of.

Technique.

So.

Let's.

Make.

A.

Let's do.

Six.

Observations.

Projecting.

On.

To.

Three.

Latent states.

With.

Two.

Actions.

For.

Fifteen.

Time.

Steps.

With.

A.

Planning.

Horizon.

Of.

Five.

Okay.

Okay.

Here.

First.

What it does.

Is it.

Does.

A.

Sequential.

Test.

Suite.

Where.

It.

Starts.

A.

Testing.

Session.

Using.

The.

Testing.

Infrastructure.

Then.

It.

Goes.

Through.

And.

Logs.

These.

Sub.

Fully.

Realized.

Models.

Like.

Let's.

Build.

The.

Matrix.

And.

Test.

The.

Size.

Let's.

Build.

It.

Test.

The.

Size.

Visualize.

It.

And.

So.

This.

Helps.

Iteratively.

Building.

And.

Coding.

As.

Well.

Or.

Let's.

Just.

Do.

One.

Belief.

Update.

And.

Then.

See.

If.

It.

Went.

The.

Direction.

We.

Thought.

It.

Was.

Going.

To.

Go.

So.

Now.

It.

Runs.

Through.

And.

Now.

It's.

Running.

The.

Full.

POMDP.

And.

Then.

It's.

Going.

To.

Be.

Outputting.

So.

Previously.

There.

Was.

Five.

Four.

And.

Three.

Five.

Observations.

Four.

Latent states.

Three.

Affordances.

Which.

Is.

Why.

We.

Saw.

That.

Trifurcation.

In.

The.

Policy.

Rollout.

Visualizations.

Now.

Here.

Here.

We.

See.

Policy.

Bifurcation.

Because.

There's.

Two.

Affordances.

Per.

Time.

Step.

So.

Here.

We.

Have.

Low.

And.

High.

Of.

Low.

Low.

Low.

High.

Low.

And.

So.

This.

Is.

A.

Way.

To.

See.

The.

Expected.

Free.

Energy.

And.

The.

Components.

Of.

Expected.

Free.

Energy.

For.

A.

Whole.

Policy.

Tree.

Rollout.

I.

Don't.

Know.

If.

That.

Has.

Been.
Done.
Even.
Of.

Few.
Weeks.
Ago.
And.
As.
Kind.
Of.
Expected.
The.
Ambiguity.
Rises.
Expected.
At.
Least.
In.
How.
This.
Is.
Visualized.
So.
That.
Took.
76.
Seconds.
For.
Running.
This.
Testing.
Suite.
That.
Ran.
An.
Informative.

Sequence.

Of.

Events.

That.

Began.

With.

Just.

Testing.

Whether.

The.

The.

Machinery.

Was.

In.

Order.

Then.

Went.

On.

Through.

Building.

Some.

Of.

The.

Sub.

Component.

Tree.

Operating.

Them.

For.

Short.

Periods.

Of.

Time.

And.

Then.

Working.

Its.

Way.

Up.

To.

Ultimately.

Something.

That.

Is.

Outrageously.

Useful.

For.

The.

Active.

In.

France.

Which.

Is.

Holographic.

Representations.

Of.

Policy.

Tree.

Rollouts.

With.

Defined.

Epistemic.

And.

Pragmatic.

Value.

Components.

And.

Their.

Balance.

So.

That's.

A.

Generic.

POMDP.

And.

Then.

To.

Not.

Leave.

Obsidian.

Out.

In.

The.

Cold.

Here.

The.

Read.

Me.

Here.

The.

Read.

Me.

Is.

Written.

Like.

Software.

Style.

But.

We.

Could.

Improve.

It.

Like.

That.

So.

Then.

We.

Have.

A.

Clear.

Link.

Going.

From.

Well.

Here's.
The.
Read.
Me.
For.
How.
The.
POMDB.
Is.
Applied.
In.
That.
Folder.
And.
Then.
Here's.
The.
Bigger.
Concept.
A.
Matrix.
It's.
Like.
What.
About.
The.
A.
Matrix.
Oh.
Yeah.
It.
Defines.
The.
Mapping.
Between.
Hidden.
States.
The.

S.

Space.

And.

Observations.

The.

O.

Space.

What's.

The.

S.

Space.

Oh.

Yeah.

It.

Looks.

Like.

That.

So.

It.

Is.

Just.

Tracing.

Around.

All.

Of.

These.

Different.

Aspects.

Of.

The.

Total.

Semantic.

Space.

But.

We've.

Juxtaposed.

Here.

Docs.

Which.

Are.

The.

Ultra.

Implementational.

Or.

Meta.

Implementational.

Examples.

And.

Concepts.

For.

This.

Repo.

As.

Repo.

Then.

There's.

The.

Knowledge.

Base.

Ranging.

From.

The.

Statistical.

Mathematical.

Foundations.

On.

Through.

The.

Agent.

Level.

Documentation.

And.

Including.

Both.

Technical.

Definitions.

Like.

What.

Is.

A.

B.

Matrix.

Technical.

As.

You.

Want.

It.

To.

Get.

And.

Then.

What.

Is.

An.

Educational.

Curriculum.

For.

Every.

Single.

Possible.

Of.

This.

For.

Every.

Single.

Possible.

Of.

That.

And.

Then.

Through.

The.

SRC.

Dispatch.

Center.

We.

Can.

Have.

These.

Implementations.

That.

Can.

Be.

Used.

In.

A.

Super.

Standalone.

Way.

Or.

In.

A.

Really.

Integrated.

Way.

So.

Let's.

Just.

Look.

At.

Few.

Of.

The.

Other.

Implementations.

This.

Is.

Also.

A.

Kind.

Of.

Generative.
Model.
That.
I've.
Never.
Built.
Before.
And.
It's.
In.
The.
Continuous.
Time.
Setting.
Okay.
So.
This.
One.
We're.
In.
The.
Continuous.
Time.
Setting.
And.
Let's.
Get.
The.
Test.
Started.
So.
Running.
The.
Test.
Suite.
In.
The.
Continuous.

Time.

Setting.

Let's.

See.

What.

It.

Looks.

Like.

Over.

In.

Obsidian.

Continuous.

Generic.

So.

Okay.

Here's the continuous generic model.

So.

Just like that.

Generic.

POMDP.

Was.

For.

Discrete.

Time.

Discrete.

State.

Spaces.

Partially.

Observable.

Markov.

Decision.

Processes.

Generalizations.

Including.

Planning.

And.

Other.

Cognitive.

Phenomena.

Hashtag.

Cognitive.

Phenomena.

Here.

Is.

Continuous.

Time.

Continuous.

State.

Space.

Active.

Inference.

Agents.

The.

Agent.

Learns.

And.

Operates.

In.

Continuous.

Domains.

Using.

Modern.

Deep.

Learning.

Techniques.

While.

Adhering.

To.

Active.

Inference.

Principles.

So.

It's going to have.

These.

Techniques.

And.

Again.

Because.

It was in a little.

Code.

Corner.

The.

Read.

Me.

Reads.

More.

Like.

A.

GitHub.

Read.

Me.

Than.

A.

Obsidian.

But.

We.

Can.

Quickly.

Evolve.

That.

By.

Saying.

Also.

Write.

A.

Mega.

Obsidian.

Style.

Densely.

Linked.

Comprehensive.

Continuous.

Time.

Model.

Here.

Connecting.

To.

And.

I'll just drag in.

Docs.

A whole folder.

And.

Knowledge Base.

So.

Let's see what it.

Writes there.

But.

Meanwhile.

Back in the test.

Lance.

So.

Initialization.

Shift.

Operation.

Sensory.

Mapping.

Just the perceptual.

Sensemaking.

Elements.

Of the continuous.

Time.

Situation.

Then.

One.

Time.

Step.

Of.

Taylor.

Series.

Expansion.

One.

Free.

Energy.

Computation.

One.

Belief.

Consistency.

Check.

And.

Then.

Several.

More.

Sophisticated.

Tests.

Leading.

On.

Through.

Ultimately.

To.

Taylor.

Expansions.

In.

Generalized.

State.

Space.

So.

Here.

The.

Model.

Has.

I'll just.

Pick some.

Random.

Variables.

This might.

Slow down.

The.

The.

Computation.

Vastly.

Three.

Latent.

Two.

Latent.

States.

Three.

Observations.

Four.

Orders.

For.

The.

Generalized.

Coordinates.

Point.

Oh.

One.

Time.

Step.

For.

Integration.

Let's.

Rerun.

The.

Tests.

So.

That.

Took.

Ninety.

Four.

Seconds.

To.

Run.

The.

Test.

Suite.

With.

That.

Previous.

One.

So.

Meanwhile.

Over.

In.

Continuous.

Time.

Active.

Inference.

Continuous.

Time.

Inference.

So.

Here's the Obsidian style.

Article.

That it wrote.

On.

What we asked it to write about.

So let's check out the neighborhood of that article.

So.

Here's our big.

Knowledge network.

And see if we can find.

And there's the red dot.

So here's continuous time.

Active inference.

And here's how it's Obsidian linked.

To all these topics.

Continuous time.

Active inference.

Free energy principle.

Okay.

Okay.

Alright.

Now we'll start the tests.

But meanwhile.

Let's look at the outputs.

From the previous.

Continuous time.

Testing.

Okay.

I'm not even sure.

What these may yield.

Okay.

Some.

Pre-flight checks.

On sensory.

Mapping.

How accurate was just.

Our.

Input versus output.

Was there even.

The right.

Correlation.

Direction.

For.

Sensory mapping.

Shift operate.

I don't know what these numbers mean.

In the JSON format.

But.

You can.

Ask.

Please improve the tests.

Of continuous.

Time agents.

To.

Yield.

Many more.

Visualizations.

Okay.

But.

While we're looking at the previous versions of the code.

Okay.

Space.

Predictions.

Now.

Maybe this was just.

When.

When there was only one or two orders for the model.

We could have just been fitting.

Free energy.

Generalized coordinates.

To.

A point.

Which.

Might be overkill.

Maybe we could have just identified the point.

But.

Some of these visualizations.

Like.

Being able to identify.

The gradient following.

And the solenoidal.

Isocontour following.

In a continuous face space.

Okay.

This gif.

Looks like.

It opened for a second though.

It was just showing.

Well.

Let's see.

Okay.

That took 104 seconds.

So we.

We slightly.

There's this gif.

It was being rewritten.

While we were looking at it.

Okay.

So.

Maybe.

It was just a weird.

Corner.

Of.

State space.

Maybe there's some.

Miss implementation.

Of something.

Maybe it doesn't.

But.

You know.

We're.

We're in the ballpark.

Harmonic motion.

Okay.

Maneuvers.

In state.

And phase spaces.

Taylor series.

And.

Taylor series.

Expansions.

So this looks like.

It's doing a two order expansion.

With a position.

Which is like.

The first moment.

Of the Taylor series.

And then the first derivative.

The velocity.

And.

Again.

Maybe these numbers are.

Hard coded.

Maybe the implementation is not perfect.

Maybe it's not general.

C'est la vie.

So on.

But.

The infrastructure is in place.

For building these things.

Okay.

Now.

The test suite.

Has been.

Graciously added.

With these several hundred.

Lines.

We don't even need to.

Type it.

In.

We could have said.

Continue to.

Run that by yourself.

Okay.

I'm going to just.

Briefly.

Show the other things.

The biofirm.

And the ant colony.

And then.

Open to.

Whatever.

Ideas and questions.

People have.

In the chat.

For.

Which.

Obsidian.

Neighborhood.

Should we build out.

Which.

Things.

Should we build.

Or coding.

Like.

Here.

Okay.

First.

Ant colony.

This one.

Was almost in one shot.

I just said.

Write a read me.

And then fully.

Totally.

Actually implement.

An ant colony.

Active inference.

Multi-agent simulation.

So.

The kind of thing.

That.

That.

Was.

Years.

Of.

Wheel spinning.

In.

2017.

18.

19.

2021.

Is.

Being.

Semi.

Autonomously.

One-shotted.

By.

Just.

A hope.

And a dream.

Let's just see.

What it.

Output.

I don't know.

If.

It has.

Run.

Basically.

It spins out.

The schema.

For a whole.

Multi-agent.

Simulation.

Now.

Is this going to be.

Compliant.

With.

Insert your own.

Favorite.

Multi-agent.

Framework.

No.

Not necessarily.

However.

If.

In the documentation.

We had.

A documentation.

For.

That.

And it was linked.

To multi-agents.

As.

One of the ways.

To do it.

Or the way.

To do it.

Then.

It would be.

Structured that way.

Let's just see.

If it's already.

Operative.

Okay.

Ant colony.

We'll run.

Setup.

Just to see.

If it needs that.

Okay.

Maybe it needs.

Commands.

Let's just see.

If simulation works.

Okay.

It's running the tests.

For the continuous.

Generic.

Case.

Over off to the side.

But it's like.

And then.

Already you can see.

How.

What I'm doing.

And showing.

With only.

One and a half.

Autonomous agents.

At any given time.

It's already.

Like the farmers market.

Local.

Organic.

Low throughput version.

Because.

Why not.

Just have this.

Dispatching.

Truly programmatically.

Without a graphical interface.

Or why not have it.

Um.

In any number of other.

Pretty adjacent ways.

Using the computer operator.

Using just.

Uh.

Headless machine.

Using just.

Uh.

GitHub.

Agents.

So many ways to do it.

Okay.

So now it's.

It is proceeding.

Through the extended.

Test suite.

With the continuous time setting.

But let's just pivot over to the ant colony.

To see if we can get some ant gifs on screen.

Ensure that.

Running this.

Really does.

Fully.

Implement.

The total.

Ant.

Colony simulation.

Yielding.

Sophisticated analysis.

And many.

Hilarious.

Dot gif.

So.

We'll see where it gets on the ant colony.

But meanwhile.

Over in the biofirm land.

So.

Biofirm.

I.

In one sentence.

Summarized our work.

On the biofirm project.

And said.

Write a readme.

And then spin out.

Totally according to how it could be.

Make it happen.

Something's like that.

So.

It's.

A homeostatic control model.

Biofirm.

Using active inference principles.

For bioregion management.

This implementation.

Demonstrates how active inference.

Can be used for maintaining system stability.

Through partial observations.

So.

The biofirm.

Implements a POMDP.

Building.

Copiously.

Off of.

Of course.

The simple.

And the generic.

Things.

Already in the adjacent folder.

And all the.

Knowledge.

Docs.

That we already looked at.

It's going to use a POMDP.

To do this.

Bioregional.

Homeostasis.

The model.

Aims to maintain system.

In the homeostatic range.

By.

One.

Making observation.

From a simplified.

Three-state observation space.

Or.

We could.

Combine.

Homeostatic.

Bioregional.

Nested stewardship.

With.

The generic.

POMDP.

So then it could be.

A different observation space.

But just to fix it.

For.

This current.

Time.

Two.

Inferring.

The true underlying.

Five-state environment.

So here.

And meanwhile.

Ant colony.

Autonomous.

Operation.

Ongoing.

Here.

There's three observations.

We're getting.

Like.

We're getting.

High.

Medium.

Low.

From.

The thermometer.

And then there's.

Five.

Latent states.

So it could be.

Off.

One end.

Or off.

The other end.

And then there's also.

These sort of.

Threshold regions.

With.

On the lower.

And the upper bounds.

You know.

Or you could have.

Five observations.

And three.

Latent states.

That's why we need.

These plain text formats.

And that's why we've been.
Developing them for years.
Because.
It isn't enough.
To just say.
Well.
Let's put.
At three.
Again.
You may be more than.
Right.
In a given situation.
But.
The bigger picture.
Is doing.
Meta.
Parameter.
Sweeping.
Across.
Configuration.
Spaces.
For.
Even.
What.
One.
Agent's.
Cognitive.
Model.
Is.
Structurally.
Let alone.
Quantitatively.
So.
Three.
The biofirm.
Selects.
Then.
Actions.

From a set.

Of three.

Decrease.

Maintain.

Or.

Increase.

To keep.

The system.

In the desired.

State.

So.

It continues.

To go through.

Here's.

Some of the.

Technical.

Elements.

Let's.

Read this.

Read me.

In.

Obsidian.

Obsidian.

So.

Similarly.

This one.

Could be.

Enriched.

With visualizations.

And all of that.

Make it a little bit less.

Like.

Code.

Read me.

A little bit more.

Obsidian.

Formatted.

And.

What it.

Runs.

Let's.

Look at some of the outputs.

From the.

Biofirm.

Okay.

Granger.

Causality.

Transfer.

Entropy.

On network.

Convergent.

Cross mapping.

Not sure what this one would be.

Causal impact.

Not sure if it's calculated accurately.

Or in any specific way.

But.

This brings us back.

Back to the dispatch operator.

Like.

Causal impact.

Doesn't.

Just have one formalism.

That's why there was a.

Great.

Work.

In a live stream.

With like the 18.

Different.

Different.

Definitions of surprise.

So then.

All of those.

Can be held up.

And then there's a dispatch.

That mediates.

The semantic use.
Of the term.
Surprise.
And might.
Calculate all 18.
Or maybe.
Seven of them.
Only apply to discrete.
So if it's continuous.
Then.
They're just not going to be.
Calculated.
Or.
There's some other preference.
Or you pick a random one.
Or you pick a different one.
Or you.
Can pass it as an argument.
Like.
Which surprise concepts.
Are we working with?
But.
All of that.
Is only plausible.
Within this space.
Where we have the.
Dis.
Hyphen.
Intermediation.
Of the.
Dispatch.
Caller.
Okay.
Causal.
Intervention.
Restoration.
We can look at the logic.
Of what these simulations do.

Okay.
Those look all like.
The same kind of plots.
Just with different.
Different interventions.
Possibly.
Okay.
Fractal active inference.
Detrended.
Fractal analysis.
DFA analysis.
Hurst exponents.
Looking at the roughness of a path.
Looking at correlation.
Scale-free correlation dimensions.
Looking at.
The fractal power spectrum.
Intervention management.
In the fractal setting.
This looks like.
Still mainly in.
Time.
But maybe this is spatial.
Then.
Here's free energy terms.
Again.
Maybe.
Maybe.
Today.
This isn't exactly.
The perfect.
Representation or calculation.
But.
Already.
With a self-improving.
Dispatch agents.
And operator.
Were.

Wildly further.

Than we were.

A few weeks ago.

Okay.

Hierarchical.

Active inference.

We could.

Say.

Make an obsidian file.

That represents.

The topology.

Of.

The biofirm.

Simulation.

Okay.

Information.

Interventions.

On networks.

Let's add.

Like.

An earth.

Plotting.

Component.

Or something.

But it just.

Okay.

That one looked interesting.

Okay.

So.

Some.

Some.

Simulation.

Some.

Oscillatory.

Simulation.

Is.

Happening.

Across.

Three.

Scales.

Interaction.

Efficacies.

And.

Strengths.

This one could be like a Pokemon character.

We could evaluate.

Like.

Farmworks.

Different kinds of.

Interventions.

And treatments.

And.

And.

Contexts.

These are attractor.

Landscapes.

We can.

We can dig down.

We said.

For every single method used.

In the biofirm.

Write.

A.

Mega.

Amazingly linked.

Article.

With obsidian format.

Okay.

Briefly.

Sidebar.

Over with the ant.

So.

It got 25 calls deep.

And then.

Cursor.

Please ping us at.

Hi.

At cursor.com.

If you think we should increase this limit.

I would like to imagine the limit being increased.

If you need more tool calls.

Please give the agent feedback.

And ask it to continue.

How relatable.

Let's just run it from the terminal again.

Fix this.

Comprehensive.

Okay.

So.

Now.

While.

Dr. Claude.

Is working on the ants.

Let's continue with the biofirm.

Okay.

Eigenvalues.

For stability detection.

On dynamical landscapes.

Valid.

Relevant.

Analyses.

Okay.

Looks like a vector field representation.

Are these points accurate?

They look pretty similar.

But.

Maybe we were in a.

Configuration set.

Where.

They went similar.

Okay.

And then this.

Is almost at the end of the outputs.

From the biofirm.

Wow.

Wow.

Wow.

If we are to take a plain.

Reading of this.

We have.

Heat map.

Coded.

Intensities.

Blue.

Low.

Yellow.

High.

Then.

In four.

Semi.

Discretized.

Chapters.

Across these.

120 scales.

Through time.

There are these.

Repeating.

Diffraction.

Patterns.

Like.

Something.

Is happening.

Very.

Optical.

Illusion.

Like.

It's just a straight line.

Across here.

But when I.

Look at it.

It looks like a sine wave.

And.

Wavelet.

Power band.

Analyses.

So.

The kinds of.

Background.

Questions.

That somebody might have.

Like.

How does.

Variational.

Free energy.

Come to matter.

In this.

Biofirm.

Implementation.

That's.

A great.

Generative AI.

Direction.

Venture.

And then there's also.

All of these.

Development.

The vibe.

Coding.

All these different.

Development.

Directions.

Like.

Now.

Please.

Do this.

For this.

Region.

Or.

Please.

Now.

I have.

This many.

Of this.

Sensor.

And this.

Many.

Of this.

Actuator.

So.

That's.

Where the.

Biofirm.

Got.

I.

Played.

Around.

With this.

Biofirm.

Specific.

Active.

Dispatcher.

So.

Let's just.

Take one.

Quick.

Look.

At.

What.

The.

Dispatch.

Looks.

Like.

Here.

We can.

Use.

Different.

Inference.

Methods.

So.

We might.

Want.

To.

Sample.

From.

A.

Distribution.

But.

Even.

From.

The.

Very.

Same.

Distribution.

We.

Might.

Want.

To.

Also.

Do.

A.

Variational.

Belief.

Update.

Depending.

On.

What.

We.

Had.

At.

Hand.

At.

A.

Moment.

So.

Again.

That.

Is.

Where.

This.

Dispatch.

Dis.

Intermediation.

Comes.

Into.

Play.

Where.

It.

Let's.

A.

High.

Level.

Understanding.

Of.

Active.

Inference.

Like.

Reflected.

By.

The.

Ontology.

Be.

Applied.

Through.

The.

Strong.

Engineering.

Of.

How.

The.

Dispatch.

Maps.

From.

The.

Ontological.

Semantics.

On.

Through.

The.

Implementational.

Pragmatics.

So.

Here.

Are.

These.

Different.

Dispatch.

Calls.

And.

This.

Is.

Again.

Just.

To.

Keep.

It.

Local.

With.

An.

Active.

Inference.

Dispatch.

Template.

But.

As.

More.

And.

More.

Things.

Get.

Made.

And.

Tested.

In.

Open.

Source.

Explored.

You.

Can.

Imagine.

That.

In.

The.

SRC.

It.

It's.

Possible.

To.

Work.

Out.

Some.

Very.

Generic.

Dispatch.

Methods.

Okay.

So.

Um.

In.

The.

Last.

Part.

Of.

This.

Stream.

We.

Will.

Sort.

Of.

Halfheartedly.

See.

If.

We.

Can.

Get.

This.

Mega.

Ant.

Colony.

Simulation.

Going.

As.

It.

Continues.

To.

Spiral.

And.

Self.

Talk.

But.

Meanwhile.

Let's.

Push.

GitHub.

Update.

Take.

A sip.

Water.

And.

Then.

If.

Anybody.

Has.

Any.

Questions.

Or.

Ideas.

Definitely.

Write.

Them.

In.

The.

Live.

Okay.

Let's.

Make.

A new.

Thing.

Path.

Network.

Okay.

Okay.

I'll write out the prompt that I eventually want to use.

We'll see if the ants.

Agents.

Get.

It's where it needs to be in the next minute while writing this prompt.

Uh.

Then.

Let it finish.

And then we'll dispatch.

This prompt.

To see if we can one shot.

A thing.

That.

Does something.

So.

Okay.

See.

See.

This part.

Of.

The.

The.

The.

The zero lag.

Okay.

Milliseconds of lag.

Between.

Applying the edits.

On a whole folder.

And then running it again.

And then instantly picking up with error.

So.

That's like where.

Vibe coding is about the vibes.

For us.

Humans.

Ants.

But.

It's a different vibe.

For.

This.

Operator.

Over here.

So.

Okay.

Here we are in path network.

We want a simulation.

Where a.

An.

Arbitrary.

Topology.

Randomly.

Initially generated.

Okay.

Here we go.

Did we get there with the ants?

25.

We hit the limit again.

Do we.

Do we roll the die one more time?

It's tempting.

Fix this.

Comprehensively.

And totally.

Run the ant simulation.

Okay.

Let's.

Let it.

Just one more.

And then.

Let's write out what we want for the path network.

Meanwhile though.

For anyone watching live.

Write any ideas or thoughts.

For.

What we could do for the obsidian.

Knowledge.

Element.

Or.

For some kind of active inference simulation.

We could do anything.

We could front run.

We could back run.

We could back fill.

Front fill.

Any of those.

We want a simulation.

We want a simulation.

Where an arbitrary topology.

Randomly initially generated.

Of nodes.

Reflecting their.

Communicative.

Effective.

Causal.

Mutual influence.

Is generated.

And simulated.

Each node.

Is a.

Continuous time.

Active inference.

Agents.

Different.

Nodes.

Have.

Can.

Have.

Different.

Relations.

With the.

Same.

Underlying.

Inference.

Methods.

For example.

Different.

Time.

Deltas.

Of integration.

Different.

Levels.

Of.

Taylor.

Series.

Expansion.

For.

Generalized.

Coordinate.

Coordination.

Inference.

Anticipatory.

Inference.

At a big picture level.

It is like.

There.

Are.

Nested.

Sine.

Waves.

That.

Are.

The.

Level.

Of.

Water.

In.

The.

World.

All.

Of.

The.

Nodes.

Are.

Like.

Ships.

Or.

Towers.

That.

Are.

Doing.

This.

Real.

Time.

Inference.

On.

How.

Much.

To.

Ascend.

Or.

Descend.

In.

Order.

To.

Stay.

With.

In.

Tolerable.

Limited.

Range.

Of.

The.

Overall.

B.

Level.

Which.

Floats.

All.

Bones.

So.

We're.

Looking.

For.

This.

Absolute.

Total.

Simulation.

We.

Know.

You.

Can.

Do.

This.

We.

Expect.

You.

To.

Do.

This.

All.

In.

Iterively.

Thoughtfully.

One.

Shotting.

This.

Whole.

Issue.

With.

Initially.

Writing.

A.

Markdown.

Obsidian.

Obsidian.

Obsidian.

Style.

Schema.

Centerpiece.

And then.

Actually.

Comprehensively.

Proceeding.

From.

There.

Given.

The.

Arc.

Of.

Development.

Outlines.

In.

That.

Very.

Same.

Initial.

Markdown.

Mega.

Prospectus.

Okay.

And.

Simulation.

Another.

Day.

New.

Agent.

Okay.

Okay.

Make it happen.

Thank you.

Okay.

Here we go.

Okay.

All right.

While.

That.

Is.

On.

Slow.

Mode.

We'll.

Keep.

Watching.

It.

From.

The.

Side.

Okay.

So.

Here.

Initially.

As requested.

It.

First.

Lays out.

A file structure.

For this.

Package.

Sub package.

Informed.

Package.

Core components.

Okay.

Continuous time.

Active inference agent.

Network topology.

Environmental dynamics.

Inference methods.

Okay.

Phase one.

So.

Made the requirements.

Text file.

Now.

Phase one.

Like.

It.

Is.

Moving.

Quickly.

Through.

These.

Phases.

Let us.

Briefly.

Return.

To the.

Live stream.

Top level.

Okay.

We.

In this agenda.

Did talk about.

Active inference.

We talked about.

Free energy principle.

A little bit.

And gave.

Ample.

Cross linkings.

And applicable.

Strategies.

For those.

Wanting to model.

Within.

We talked.

In terms of.

The things.

About.

The continuous.

Generic.

Model.

And the generic.

POMDP.

There's also.

A simple.

POMDP.

Which is.

Let's just.

Quickly.

Look at the output.

This is just.

For learning.

The basic.

POMDP.

This.

Is.

A useful.

Learning.

Resource.

And it just.

Gives.

A.

Example.

Of.

An active.

Inference.

POMDP.

Agent.

That you can.

Ask.

Questions.

About.

And all of that.

It doesn't.

Use any other.

Packages.

Or anything.

Else.

Like that.

Okay.

And there was.

The generic.

One.

Then there was.

The ant.

Colony.

Simulation.

That our.

Agents.

Was.

Still.

Repeatedly.

Hitting.

His head.

Against the wall.

25 or more.

Times against.

Then there's.

The biofirm.

Which was doing.

The active.

Inference.

Fractal.

Okay.

Codevis.

Let's do a.

Quick.

Codevis.

With our.

We have one.

Free flow.

So.

Okay.

Okay.

Let's.

Say.

Given the.

Totality.

Of this situation.

What.

Would you.

Honestly.

Comprehensively.

Say.

About.

The topology.

And geometry.

Of the.

Package.

Okay.

We'll.

See what.

Codevis.

Does here.

We talked.

A lot.

About.

The.

POMDP.

And.

The matrices.

Looked.

At.

Some.

Of the.

Implementations.

For.

The matrices.

Talked.

A little.

Bit.

About.

The testing.

Paradigm.

And.

The runner.

And.

The dispatch.

Operator.

Paradigm.

Practical.

These.

Examples.

I guess.

Are places.

Where it.

Could be.

Applied.

But.

They.

Weren't.

Applied.

Today.

Visualization.

Methods.

We saw.

A bunch.

Of.

Visuals.

Future.

Directions.

Extended.

Functionality.

Performance.

Optimization.

Additional.

Test.

Cases.

Documentation.

Improvements.

Community.

Engagement.

Contribution.

Guidelines.

Issue.

Tracking.

Feature.

Requests.

Collaboration.

Opportunities.

Okay.

Repo.

Information.

Next.

Steps.

References.

And.

Notes.

So.

Jump.

Through.

That.

Okay.

Here's.

Well.

Let's look at first.

CodeViz.

While.

Okay.

Path.

Network.

Looks like.

It's basically done.

So.

Let us.

Okay.

It still.

Is generating.

CodeViz.

Let's see.

What happens.

With.

Path.

Network.

Okay.

Okay.

Just running.

The.

Requirements.

Installation.

Script.

Okay.

Looks like.

Either.

It's not going to work.

Or it will work.

And.

The example.

Okay.

Oh.

Python 3.

Okay.

Okay.

Comprehensively.

Continue.

To.

Fully.

Develop.

The.

Example.

So.

That.

Running.

That.

One.

Line.

In.

Terminal.

Alone.

Fully.

Logs.

Totally.

And.

Outputs.

Safely.

All.

The.

Visualizations.

And.

Okay.

Andrew wrote.

Broad.

Intentionally.

Semi-vague.

Question.

What do you think about using your repo for generating simulating environments?

You've already spoken on it, but what's next?

What's still obscure?

Good question.

So.

It still takes its time.

I think that it will be interesting.

This is not in response to the question, but just.

Obsidian link is.

The topology of the Obsidian network is only strictly which use the Obsidian linking.

The double bracket format.

So.

That's.

Great.

It's also slightly different than a hypertext link.

It's slightly different than other kinds of links that might exist.

So.

It's not the only way to link, but.

It's cool that.

It's so amenable to cursor's creativity.

And.

It could be in a more advanced setting.

Flipped out.

With other.

Plain text benefits.

Or other linking.

Paradigms.

So.

Just to say that CodeViz.

Which I am excited for the development of.

I believe it offers a contrasting.

Possibilities.

For.

Structural and topological investigation of complex knowledge bases.

And I'm really curious to see.

How it goes.

How it goes.

With their development of labeled edges.

And some other features.

Because.

It could give us these kinds of systems diagrams.

Hopefully.

That would complement.

Different kinds of static analysis.

And knowledge base style linking.

Amongst componentry.

So.

Two files might not be.

Linked to each other through Obsidian.

But they might be very, very related.

From an actual.

Functional perspective.

So.

If.

Those kinds of.

Hidden functional relations.

Can be.

Inferred.

With.

CodeViz.

And then.

Meanwhile.

We could use.

Plain text style.

Strategies.

For knowledge base linking.

And.

Change among plain text formats.

It would be.

Wonderful.

Okay.

Let's see where we can get with the path network.

See where we might get with the environment.

And then see if anyone else has funny.

Or interesting ideas from.

Live chat.

Okay.

Okay.

Looks like now.

It uses a.

Shell script to run.

Let's just look at it first.

Sets up a Python virtual environment.

Installs dependencies.

Runs the simulation.

Let's just see if we can run the simulation by itself.

Okay.

All right.

Okay.

While agent is fixing the simulation.

Yeah.

Generating and simulating environments.

I think here.

We have a lot of fun.

Research and work.

I guess.

My.

Broad.

Intentionally.

Semi-vague.

Question is.

Are we not just talking about the other side of the coin?

I mean.

Single or multi-agent design.

Is like.

What we've chosen.

To.

Ladle out.

From the total scenario.

And.

If we're going to be talking about.

What we've ladled out.

And what we have left.

Unladled.

And we're looking at it from both sides.

And perspective swapping.

Then the difference between.

Environmental modeling.

And agent modeling.

Is going to.

Blur.

Observer.

Yet still remain.

In practice.

Totally precise.

But.

For.

Alice and Bob.

Interacting.

Across.

Across the interface.

Whether.

A third party observer.

Or one or the other.

Asserts themselves.

To be.

The environment.

Or the agent.

Or they both think they're.

This one or that one.

Or they both think the other is that.

Or the other.

I don't think from a.

Category theory.

Slash.

Systems engineering.

Slash.

Markov blanket perspective.

It's going to matter.

Too much.

Implementationally.

So.

Rhetorically.

Or for a given.

Audience.

Or domain.

Their.

Entire.

Understanding.

May hinge.

On what the agent.

And what the environment.

Is.

What scale.

What side of the wall.

And all that.

However.

That wouldn't.

Necessarily.

Break any.

Symmetries.

Regarding.

The.

Formalism.

Okay.

And.

Now.

We're back.

With the.

Path network.

Clear it.

Run the example.

Okay.

This is a common error.

It will often use.

Like a certain visualization.

Backend.

Initially.

That.

Doesn't work on my computer.

It doesn't work on Linux.

Or something like that.

And then.

It figures out.

Oh.

We're going to use this other.

Visualization.

Backend.

But.

That's where.

Having this mega documentation.

So it's like.

When I say visualize.

Here's what I mean by that.

Or when I say.

Make it like a GIF.

Here's the.

Here's some good code.

For doing that.

So then instead of.

Being.

Loosely.

Associatively.

Dipping into.

The LLM.

Semantic space.

There's an.

In context.

Option.

For.

Transfer learning.

Okay.

Let's just.

Roll the die again.

And run it cleanly.

Thank you for the question.

Andrew.

Gomez.

Thank you for joining.

Appreciate that.

XEQ.

Thank you.

Hello to you as well.

Okay.

Let's see what it.

Output.

Wow.

Okay.

15 agents.

Looks like they all converged.

To this height.

Network topology.

Water level distribution.

So this is the occupancy.

Histogram.

Of what.

Water.

Of what.

Through the.

Simulation's duration.

Where it was at.

Time.

So.

As.

Those.

Watching live.

Or re-watching.

Can see.

We started.

Just a couple minutes ago.

With that.

Kind of.

Long.

Esoteric.

Ish.

Prompt.

And then.

We just had to.

One or two times.

Just say.

Fix it.

And then on the third try.

We're totally in the game.

Now.

Comprehensively.

Improve this.

Have the.

Have a configuration.

File.

At top.

Level.

Of folder.

For.

Easy.

Modification.

Ensure.

There are.

Many.

Kinds.

And.

Comprehensive.

Approaches.

To visualization.

Including.

Animations.

Dot.

Skip.

Okay.

XZQ.

Wrote.

What's the game?

Okay.

Environment.

path network looks through how it's going to implement it. Okay, we have number of nodes, connectivity fraction, weighted connectivity, dynamic topology true or false,
topology update interval. Then we have, by interesting analogy, possibly to the micro,
meso, macro three-layer nested simulation from the biofirm, we have these three nested sine wave components. These are like three hierarchical dynamical continuous components.
And then total time steps, save intervals, agent parameters.
For a sine wave, Taylor approximation of two might be totally enough because basically just following the gradient. But then you could imagine, depending on the inertia of the agents,
maybe it can't turn fast enough. And so that's interesting, like which of the small, medium,
and large sine waves will, depending on the delta T, the step integration that each agent does,
there's so many interesting dynamics. Again, so many that kind of the move is these plain text generative formats because they are wildly deterministically and probabilistically interactive.
Okay. Let's let it finish its process.
Yeah, it's like detecting what Linux subtype I'm using. I'll never tell you.
Okay. And that might be relevant, but again, plain text, GitHub repo, use it or whatever.
All right. So we'll delete those outputs.
Okay.

Okay.
Okay.

Okay.
Okay.

Okay.
Okay.

Okay.
Okay.
Okay.
Okay.

Okay.

Okay.

Okay.

Okay.

Got some ways along and then maybe it doesn't have this package.

Okay.

Okay.

Okay.

Okay.

Wow, though.

Yeah.

A lot of fun, hopefully useful.

Again, we'll see this is kind of real

copy all the markdown files for Obsidian and transfer it over.

We can set up that knowledge repo and we can organize it in different ways.

So for those who are like, huh, maybe this could accelerate my learning or research.

With the time that you save from that acceleration, maybe consider getting involved or contributing

because we can do a lot together and we can de-risk together in ways we couldn't alone either.

So again, if this is interesting or you think like, but I just was looking for a blog that

now I see I could just generate locally or semi-locally.

If those are interesting or feelings that you have, then I encourage you to collaborate

and get involved with Active Inference Institute and Community.

Because again, this is just ad hoc surfing the edge of cursor and my own arbitrary studies.

But with our own curiosities and expertises, we can have some information environments

that have never been seen before.

So it's taken a kind of virtual environment strategy.

I don't know if that's because it's using some graphical visualization method.

But again, not to always return to it.

That is where the disintermediation dispatch operator would come into play.

Is here's what packages I like to use for this.

Or here's how you set up a virtual environment this way.

Or in the cursor rules, never set up a virtual environment.

Always do it this way.
Or install package or don't install packages.
But then install package could be a page.
And then it could be like, here's what install packages looks like in Julia.
Here's what it looks like in Python.
Here's what it looks like in this other language.
And then those kinds of articles can just be...
You may search and never find the perfect blog explaining how package installation
is similar or different in Go, Julia, and Python.
Or choose your fourth one or whatever.
The content might be there.
It might be behind paywalls or never created.
It might be tracking and cookies.
It might be unhealthy in other ways.
It might do all kinds of weird stuff.
Not that what is output from the LLM either should have the instantaneous ring of truth.
However, it's trained on a lot of that material.
And so it gets us far in this plain text setting.
And then if something doesn't work, we'll either know or we won't.
Or we'll know when we know.
And then we can fix it in the plain text format.
So...
Wrapper.
Okay.
Let's see.
So just that you can only have one of these composers.
But the chat, even though it's very functionally similar, we can do that.
So...
Let's...
Make...
One more random thing.
Let's just do a baseball game.
Here we'll use O3 Mini just to use a different model.
Comprehensively write a simulation manifesto for a baseball game.
Active inference simulation.
Torch.
This one...
See, this one's now it's downloading Torch.

766 megabytes.

So that's where...

It's...

It's...

Some overkill is happening.

So I'm gonna just stop that one.

But...

It could happen.

Plain text.

But again, chat and non-clawed models often output their text like here.

Which is like, okay, that was...

That would have passed for amazing a month ago.

But...

We're in the point where...

Having it in the composer...

Just...

This is a useful contrast.

So O3 Mini in chat.

That's the control shift L.

It did sort of write this summary.

And action plan.

For...

The baseball...

Simulation manifesto.

But...

I think we can expect that the agent mode claw at 3.5.

Is gonna be doing a whole lot more.

Okay.

X, Z, Q, Z, Q, Z, Y wrote...

Interesting.

I note...

That the wrapper is within the hierarchical structure.

Cool.

All right.

With...

That...

I will...

Make a last...

Push.

And then...

If anybody has any last comments...

I'll read them from the live chat.

Okay.

Last push.

There's the repo.

Ending of the stream.

Thank you.

Thank you.

Thank you.

Methods, too.

Or, plain text can be processed, and you can use NetworkX, you could use Cytoscape.

But the Obsidian one is pretty sweet.

Thank you.

Brief little OBS fractal for the fractal enjoyers.

Okay.

Yes.

Also looking forward to collaborate with the Knowledge Graph and more.

Okay.

Thank you, Andrew.

Yes.

Also looking forward to collaborate with the Knowledge Graph and more.

XCQ, the play happens in the field in which the agent is outside of.

Life is a ballgame.

Till the next stream.

Enjoy the repo.

Life is a ballgame.

All right.

Thank you all.

Till the next stream.

Enjoy the repo.

Get involved.

Have fun.

Act in first serve.

Leave a comment.

See what happens in your own hands and more.

Thank you.

Bye.